

B Swagath

Davanagere, Karnataka

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Professional Summary

Motivated MCA student with hands-on experience in Python, Data Science, Artificial Intelligence, and Machine Learning. Skilled in developing data-driven applications using Python, TensorFlow, OpenCV, and SQL. Passionate about solving real-world problems and eager to contribute as a Software Developer or Data Science professional.

Technical Skills

Programming Languages: Python, Java, C, and C++ with a strong understanding of Object-Oriented Programming concepts.

Web Technologies: HTML5, CSS3, JavaScript, Bootstrap, Flask, and Django for designing responsive and dynamic web applications.

Database Management: MySQL, Database Management Systems (DBMS), SQL queries, database design, and CRUD operations.

Data Science & Machine Learning: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, OpenCV, Streamlit, data preprocessing, exploratory data analysis (EDA), predictive modeling, and machine learning fundamentals.

Developer Tools: Git, GitHub, Visual Studio Code, Android Studio, Jupyter Notebook, PyPDF2, and Microsoft Office.

Core Concepts: Data Structures, Algorithms, Object-Oriented Programming (OOP), Software Engineering, REST API basics, Software Development Life Cycle (SDLC), Debugging, Testing, and Cybersecurity Fundamentals

Internships

Python with DataScience Intern

QSpiders, Davanagere

- Worked on Python programming and Data Science projects using **Jupyter Notebook**.
- Processed and analyzed structured datasets using **Pandas**, **NumPy**, and **SQL**.
- Created visual reports and dashboards using **Matplotlib** and **Seaborn** for effective data interpretation.
- Applied machine learning techniques using **Scikit-learn** to build and evaluate predictive models.
- Performed data cleaning, feature selection, and exploratory data analysis (EDA) to improve model accuracy.
- Learned statistical analysis, model validation, and data visualization techniques for real-world business scenarios.
- Strengthened problem-solving and analytical skills through hands-on implementation of data science workflows.

Projects

1. Bank Loan Analysis

Technologies Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn.

- Developed a data analysis and predictive modeling system to analyze bank loan applications using Python and Data Science techniques.
- Performed data cleaning, preprocessing, and transformation using **Pandas** and **NumPy** to improve data quality.
- Conducted Exploratory Data Analysis (EDA) to identify trends, customer behavior, and factors affecting loan approval.
- Created interactive visualizations using **Matplotlib** and **Seaborn** to present insights and support decision-making.
- Applied Machine Learning algorithms using **Scikit-learn** to predict loan approval based on customer information.
- Evaluated model performance using accuracy and other evaluation metrics to improve prediction reliability.

- Generated analytical reports to assist financial institutions in minimizing loan default risks.
- Strengthened practical knowledge of data preprocessing, predictive analytics, and machine learning workflows.

2. AI Photo Editing Application

Technology used: Python, Streamlit, Machine Learning

- Developed an AI-powered Android photo editing application with multiple image enhancement features.
- Implemented image editing functionalities including crop, rotate, brightness, contrast, saturation, sharpness, and filters.
- Designed an intuitive and user-friendly interface for seamless photo editing experience.
- Integrated image selection and image saving functionality for efficient photo management.
- Applied image processing techniques to improve image quality and editing performance.
- Optimized application performance to ensure smooth editing with minimal processing delay.
- Tested and debugged application modules to enhance usability and reliability.
- Improved understanding of Android application development, UI design, and image processing concepts.

3. Newborn Baby Weight Prediction

Technologies Used: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Jupyter Notebook

- Developed a machine learning model to predict newborn baby weight based on maternal and clinical health parameters.
- Collected, cleaned, and preprocessed healthcare datasets using **Pandas** and **NumPy** for accurate model training.
- Performed Exploratory Data Analysis (EDA) to identify relationships between maternal factors and newborn weight.

- Applied regression algorithms using **Scikit-learn** to predict birth weight with improved accuracy.
- Evaluated model performance using regression metrics such as Mean Absolute Error (MAE) and R^2 Score.
- Visualized prediction results and feature importance using **Matplotlib** for better data interpretation.
- Enhanced model performance through feature engineering and parameter optimization techniques.
- Strengthened practical knowledge of predictive analytics, healthcare data analysis, and machine learning model development.

Education

Master of Computer Applications (MCA)

UBDT College, Davanagere | 2024 – 2026 CGPA:7.0

Bachelor of Computer Applications (BCA)

BS Channabasappa First Grade College, Davanagere | 2024 CGPA: 8.0

Certifications

- Python Data Science Internship – QSpiders
- Cloud Computing Certification (VTU)

Soft Skills

- Communication
- Problem Solving
- Team Collaboration
- Quick Learning
- Report Preparation